Forensic reliability of iPhone step count and floor data across different walking speeds on stairs

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***Abstract*—While horizontal movement registered by iPhone's Health app is well-validated, the forensic relevance of vertical movement data, such as flights climbed, remains less understood. This research investigates the reliability of vertical movement data of iPhones, by looking at floor detection via the Apple Health app when walking stairs. It specifically focuses on the influence of walking speed on the registration performance, by analysing the mean average error (MAE) of step count registration and the accuracy of floor registration. An experiment was conducted, with participants ascending and descending stairs under six conditions: walking, running and slow-walking, and for each walking speed they climbed one and two floors. Data was extracted from two iPhone models and compared against manually recorded values (providing ground truth). The results demonstrated that walking speed affects registration of steps and floors; normal walking has the highest precision, while running caused substantial under-registration of both steps and floors. Slow-walking over-registered steps, presumably due to irregular gait patterns. Notably, the study revealed that correct registration improved during two-floor trials compared to single-floor trials, suggesting that greater altitude changes promote reliable sensor readings. This research is unique in its analysis of the effect of movement velocity on vertical movement registration by iPhones. It highlights the value of this data for forensic practitioners, and provides a critical view on the limitations of its use. The findings conclude that while Health app data could assist in verifying alibis or reconstructing activities, the inaccuracies at varying speeds mean it should not be used as stand-alone evidence in legal contexts.**

***Index Terms*—Digital forensics, Apple Health App, floor registration, walking speeds, iPhone sensors.**

1. Introduction

Smartphones have transformed from simple communication devices to multi-sensor recording devices, registering step count, GPS location-data, along with timestamps. They provide a wealth of data that can be used in forensic investigations, allowing reconstructions of a subject's activities or verifying alibis. While data on plane surfaces with horizontal movement is well-validated, research on the reliability of vertical movement data, specifically floor registration-data, is still developing. [1] [2] [3].

Using a process called sensor fusion (data processing from multiple sensors simultaneously), the Apple Health app records the amount of flights climbed. A flight of stairs is defined, by Apple, as an elevation gain of approximately 3 meters.

For this it uses a barometric altimeter: a sensor measuring small changes in atmospheric pressure. As altitude increases, atmospheric pressure decreases, and vice versa. It also uses data from a tri-axial accelerometer: a sensor measuring acceleration forces in order to identify gait patterns.

For the iPhone to correctly register a floor, these two sensors must combine their data and fall within certain values. These values are not publicly available and are only known by Apple. If the barometric altimeter detects a pressure change, while the accelerometer detects a gait pattern, a floor is registered. This prevents the phone from registering floor changes during other activities, such as using an elevator [2].

In previous research, participants walked routes of known distances, and the accuracy of step counts and distances recorded by the Apple Health app was analysed [4]. The study found that step registrations were generally reliable, with a mean average percentage error (MAPE) of 1.9% to 2.3% for walking and showing a higher MAPE when running. However, walking on stairs was not studied. Furthermore, other studies have shown that the Health app records floors predominantly walking upstairs, while almost never when walking downstairs [2].

Vertical movement data can prove, in a legal context, whether or not a subject changed elevation during the time that is of relevance to the case. It is crucial, however, that this data is reliable enough to be used in court, and that its limitations are known. While some research has been done into the accuracy of floor registration using the Apple Health app [2], there has not yet been a clear look into how specific physical behaviors, such as running or walking slowly, impact the reliability of the registrations. As floor detection is dependent on movement, examining step count data can provide valuable information when evaluating the reliability of floor registration. Accurate step count registration during stair climbing can complement conclusions made about floor registration.

The aim of this research is to get quantitative data on the reliability of floor registration in the Health app, specifically for different walking paces. This could prove to be useful in court, where claims regarding elevation need to be disputed or confirmed. To guide this research, the following research question was used: **How does walking speed influence the accuracy and error margin of step and floor registration of iPhones when climbing stairs using Apple's HealthApp?**

Participants ascended and descended stairs with different walking speeds, after which step count and floor registration recorded by two iPhones were compared with manually recorded ground truth data. Reliability was assessed by calculating the mean absolute error (MAE) for step counts and an accuracy score for floor registration.

2. Methods

A. iPhones

For this research, data was gathered using two iPhones belonging to two of the researchers: an iPhone SE 2022 (iOS 26.2) referred to as iPhone SE, and an iPhone 13 mini (iOS 18.6.2) referred to as iPhone 13m. Zandwijk and Boztas

(2019) [4] brought to attention the option of personalising the biometric data, such as height, weight, age and sex, in the Health app, introducing it separately for each participant. However, during the experimental phase of this research, biometric data was not changed in the Health app user interface. The iPhone SE, has the biometric data preset for sex (female) and age (24). The iPhone 13m has no biometric data set in the app.

An important setting that has the potential to affect data registration is an iPhone's battery life. In the phone's settings a disclaimer can be found, specifying that using the device in energy saving mode limits background activity and processing speeds temporarily. To avoid any influence on data registration, the phones were charged before starting the trials and energy saving mode was turned off manually.

Some preliminary testing on step and floor registration in the Health app revealed that iPhones require a long break, between 3 and 10 minutes, when the phone is turned off for complete data registration. Due to time constraints both devices were set to never automatically lock (keeping them turned on throughout the duration of the experiments), resulting in immediate data registration.

B. Participants

This study was voluntarily conducted with four students from the master Forensic Science at the University of Amsterdam. The test group consisted of one male and three female subjects, for which the relevant testing characteristics are visualised in Figure 1. During the experiment participants wore denim jeans, and carried both iPhones at the same time during each trial.The iPhone SE was placed in the left back pocket and the iPhone 13m in the right back pocket. Back pockets were chosen, as this is a very common place for carrying ones phone. Device heights from the ground were measured to ensure repeatability and no variations between trial days.

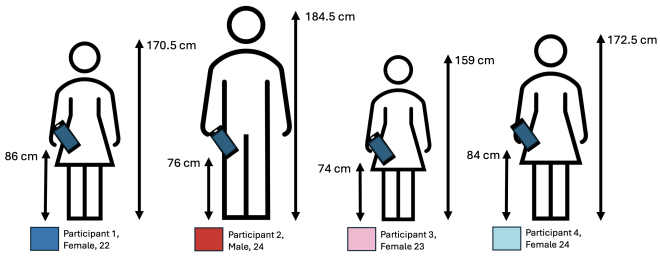


Fig. 1: Overview of participants. For each participant, their height is represented with the right vertical line, and the height of the phone in the back pocket is represented with the left vertical line. The age and sex of the participants is stated beneath each participant. The participants are represented by a color for better tracking of their individual results presented in the next chapter.

C. Experimental design

To test the accuracy of an iPhone's step and floor registration for different walking conditions, participants would either walk, run, or slow-walk. Walking was defined as the regular walking pace of the participants, running as sprinting up the stairs and slow-walking was simulated by dragging a suitcase (65 cm x 54 cm x 32 cm) of 13.3 kilograms upstairs. This weight was chosen to slow the walking pace down considerably, but not as much as to completely tire out the participants or delay the experiment. For each walking pace participants walked one and two flights, resulting in a total of six different experimental conditions.

For each separate trial the start time was noted in accordance with Central European Time (CET). While climbing the stairs with the relevant speed, participants counted their steps which would be used as the ground truth. Once the participant reached the top of the stairs the end time for ascending was noted, and step count registration by the phones was checked. When the phone had registered the step count, the participant was signaled to descend the stairs, again counting their steps. For descending the start and end time were noted down in the same way. During preliminary trials the time for registration varied between 0 and 2 minutes. Therefore, a break of 3 minutes was implemented during and between each trial for the phones to register the steps.

Once returned downstairs, the end time was noted, and the phones were checked again for registration of the step count, as well as floor data. Note that checking the phones during and after the trial was only for the purpose of knowing when to start walking again, as the phones might concatenate steps over a larger time-interval, which makes subsequent data-analysis difficult. A schematic overview of a single trial can be seen in Figure 2.

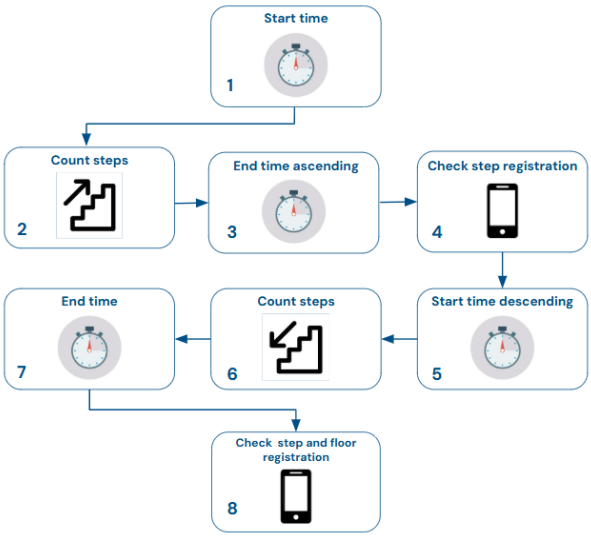


Fig. 2: Schematic overview of one experimental trial. 1) The start time was noted according to CET. 2) The participant walked upstairs, while counting their steps. 3) The end time for ascending was noted. 4) Registration of steps was checked on each iPhone. 5) The start time for descending was noted. 6) The participant walked downstairs, while counting their steps. 7) The end time for descending was noted. 8) Registration of steps and floors was checked on each iPhone.

The dimensions of the staircase used for the experiment are shown in Figure 3. The total height of the staircase was 3.88 m, and is taken as the height of one floor. This is in accordance with the standard floor height of approximately 3 meters defined in the Health app.

In each one-floor-trial a participant traversed a distance of 8.065 m. In the trials with two floors (7.76 m total height traveled vertically), the subjects traversed a total distance of 18.329 meters.

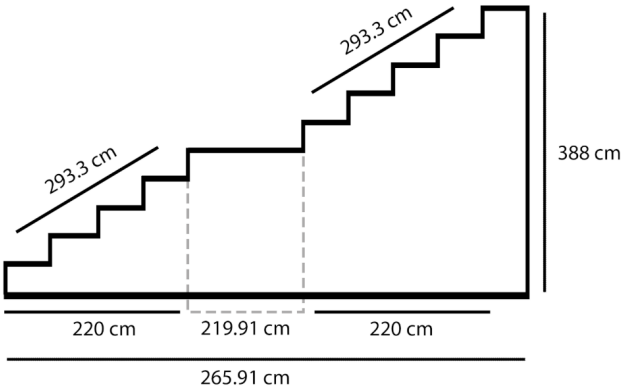


Fig. 3: Dimensions of staircase used for ascending/descending one floor. The stairs to the second floor have the same dimensions. In between the two sets of stairs there is another platform of 219.91 cm.

D. Data collection and preprocessing

It is known that an iPhone device stores digital traces on ascending and descending floors in the system files *healthdb_secure.sqlite* and *cache_encryptedC.db* [4].

1) Data extraction: To access the *cache_encryptedC.db* file a jailbroken iPhone is necessary, since it is a system level database file stored in encrypted directories. From the StepCountHistory table in the *cache_encryptedC.db* file a registration of steps, distances, ascending and descending floors can be found. Jailbroken phones were not available for use and, therefore, the *cache_encryptedC.db* file could not be obtained. For this reason, only similar data from the *healthdb_secure.sqlite* file was analysed.

The Health app database also contains registration of steps, distances, ascending floors, and other data recorded by the Health app. This database does not contain information on descending floors, only step count can be analysed for descending trials. Instead of extracting the *healthdb_secure.sqlite*, direct extraction of all Health app data from the Health app itself is also possible, using a built-in export function in the app. The directory obtained this way (*apple_health_export*) contains an export.xml file which has the same data as the *healthdb_secure.sqlite* file, but it has a different structure and is a plain text file. An example of a step count record can be seen in Figure 4, which shows a record with a startDate and endDate in which the movement has taken place with a small time granularity. The creationDate is the actual date and time when the record was stored.

The Health app data has a lot of different types of data that show records all the way back from when the Apple account was first registered on the phone. For this research only the records related to the step count and flights climbed during the trials will be used.

```
<Record type="HKQuantityTypeIdentifierStepCount" ..., creation date:2025-09-06
19:33:24 +0000&gt;" unit="count" creationDate="2025-11-10 07:00:35 +0100"
startDate="2025-11-10 06:48:09 +0100" endDate="2025-11-10 06:52:23 +0100"
value="376"/>
```

Fig. 4: Example of a step count record from the extracted Health app data. The dots (”...”) replaced additional information about the user’s phone, which was removed for privacy protection.

2) Pre-processing step and floor data: Before processing, the raw data had to be converted to usable csv-files. This was done with Python. The scripts used for this can be found on the github page that was created for this research: GitHub repository [5].

The first step was to filter the data to only include the data from the trial day and convert that to a csv-file. Next, separate csv-files were created for the step and floor count.

The phones were continuously on and the Health app was kept open, resulting in the phones constantly logging step count data every few seconds. This produced many logged rows, containing a small number of recorded steps. Which resulted in the Health app containing multiple rows of step records that fell between the start and end times of a given trial. For that reason the total number of steps recorded during each trial needed to be calculated.

To automatically calculate the total number of steps that were logged for a certain trial, code was written to sum all step entries that occurred within the manually recorded start and end timestamps of a trial. This process was then automatically repeated for all documented time intervals. There was a buffer added of 2 seconds before and after the interval since the step registration on the iPhone was sometimes delayed by 1 or 2 seconds. This data was then manually added together with the data from the trials as *registered_steps_ascend*, *registered_steps_descend*. The total number of registered floors for each trail was checked during the trials themselves, and manually written down after each trial as *registered_floors_ascend*.

E. Data-analysis

Data analysis and visualization was done with python code, which can be accessed on the github page: GitHub repository [5].

1) Step analysis: For the step registration, the mean absolute error (MAE) was used. This metric gives, on average, the amount of steps the phone was off compared to the ground truth, which is the actual amount of steps that were taken. It calculates the difference between the registered and counted steps for each trial, and then averages over the total number of trials. This reflects the actual size of the mistakes. The overall MAE is separated for both phones and broken down by the walking conditions and ascending and descending trials.

2) Floor analysis: For the floor registration, the accuracy of correctly registered floors was calculated. This metric represents the percentage of labels for which the predicted labels exactly match the corresponding ground truth labels. For this analysis, the predicted labels are the registered floors for each trial and the ground truth is the corresponding one or two floors that were walked. The overall accuracy is separated for both phones and broken down by the walking conditions and the different number of walked floors. For each of the participants, the average velocity over the trials was calculated in meters per second (m/s) to further separate the walking conditions. This was done by taking the average distance of a trail of one floor or two floors and dividing it by the average time each participant took to walk that distance. The average velocity and accuracy of each participant, for each condition and floor level, was plotted in a single graph

to further illustrate the relationship between the walking speed and accuracy.

F. Ethics and conflict of interest

In a forensic context, an individual whose phone becomes evidence in a case must be thoroughly informed of the process of seizing their personal device. Consent needs to be given for their sensitive data to be extracted and processed by the police or a digital forensics team, with the purpose of being presented in court. If necessary, a court-issued warrant could be sufficient to obtain a person's phone, solely with the purpose of analyzing case-related information.

This research is a personal, independent project, realized by the authors themselves as volunteers. Ethical approval was not sought, as it adhered to ethical standards of research, ensuring transparency in methodology and data analysis.

All volunteering students involved in the realisation of this project were fully informed about the purposes of this research and how their trial data would be used and stored. Permission to use the iPhone devices of two of the participants was obtained verbally, and they consented to their personal data being extracted and processed, within the scope of the project. A mutual agreement between the authors stated that no sensitive data would be collected, only processing trial-specific information.

The authors declare no personal or financial conflicts that could have influenced the work reported in this research paper.

3. Results

A. Calculation of velocity

To get a more precise idea about the different walking speeds for each participant, an overview of the average velocities is presented in Table I. It can be seen that the average speed varies between participants for each walking condition. Participant 3 seems to be the fastest for each of the walking conditions, where the other participants have more similar walking speeds for all conditions.

TABLE I: Average velocity of each participant for ascending and descending. The average velocity of each participant was calculated using the average walking distance and the time it took to finish the trial. The velocities were averaged over all trials and their separate walking conditions.

Participant	Direction	Average Velocity (m/s)		
		Slow walking	Walking	Running
1	Ascending	0.357	0.631	1.125
	Descending	0.380	0.698	1.243
2	Ascending	0.454	0.696	1.713
	Descending	0.527	0.929	1.610
3	Ascending	0.433	0.700	1.210
	Descending	0.408	0.714	1.121
4	Ascending	0.353	0.662	1.320
	Descending	0.326	0.674	1.440

B. Step data

1) Step registration: An overview of the results for the step registration can be found in Figure 5. The results for one and two floors are comparable for both phones. For each participant in the running trials, the steps were not registered as well as for the other two walking conditions. The difference between the maximum and minimum number of registered steps is the biggest here. Moreover, the average registered steps are always lower compared to the actual number of steps, this is 22 for one floor and 44 steps for two floors (for participants 1, 3, and 4). Sometimes the phones did not register any steps at all when running. An exception is the results for participant 2. When running up and down the stairs participant 2 took two steps at a time, so the expected number of steps is 12. The results for this participant show that the average registered steps were close or equal to the expected number of steps.

The average steps for walking are very close to the expected number of steps, although occasionally it registered either a couple of more or a couple of less steps than were actually taken.

For the slow-walking condition there is also a large range between the maximum and minimum number of registered steps. However, when slow-walking, the average registered steps are often higher than the expected number of steps.

2) Mean absolute error of step registration: To determine how often the phone was wrong in registering the steps and with what error margin, the mean absolute error (MAE) was calculated for ascending and descending. The MAE per phone per walking speed is shown in Table II, and the MAE for each participant is shown in Table III. All conditions show error in registration, and the MAEs for descending are bigger than the MAEs for ascending. Running had the largest MAE for both ascending and descending, and walking had the smallest.

TABLE II: MAE for the step count registration for each walking speed per iPhone.

iPhone	Type	MAE ascent	MAE descent
13m	slow walking	5.35	8.30
	walking	1.75	2.77
	running	11.10	17.98
SE	slow walking	5.80	8.35
	walking	1.27	3.77
	running	10.28	18.15

TABLE III: MAE for the step count registration for each participant per iPhone.

iPhone	Participant	MAE ascent	MAE descent
13m	1	5.17	13.80
	2	3.13	4.97
	3	5.20	8.13
	4	10.77	11.83
SE	1	3.53	11.53
	2	3.83	7.07
	3	5.83	10.33
	4	9.93	11.43

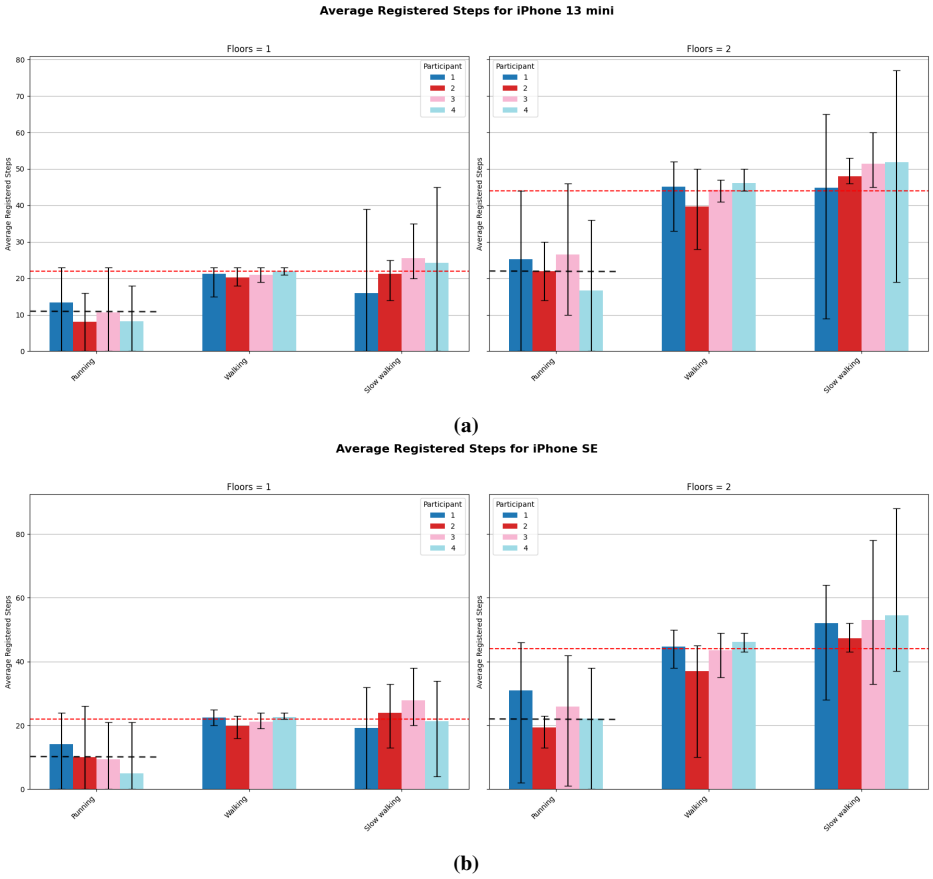


Fig. 5: Average registered steps for each condition and participants for iPhone 13m (a) and iPhone SE (b). The different conditions are shown on the x-axis (running, walking, slow-walking), and on the y-axis the average number of steps that were registered by the iPhone. Each participant is represented by a different colour. The error bars show the highest registered number of steps (top), and the lowest registered number of steps (bottom). The red dashed line, at 22 for 1 floor and 44 for 2 floors, represents the expected number of registered steps. The black dashed line at 12 in the running condition represents the expected number of steps for participant 2, as they took less steps when running.

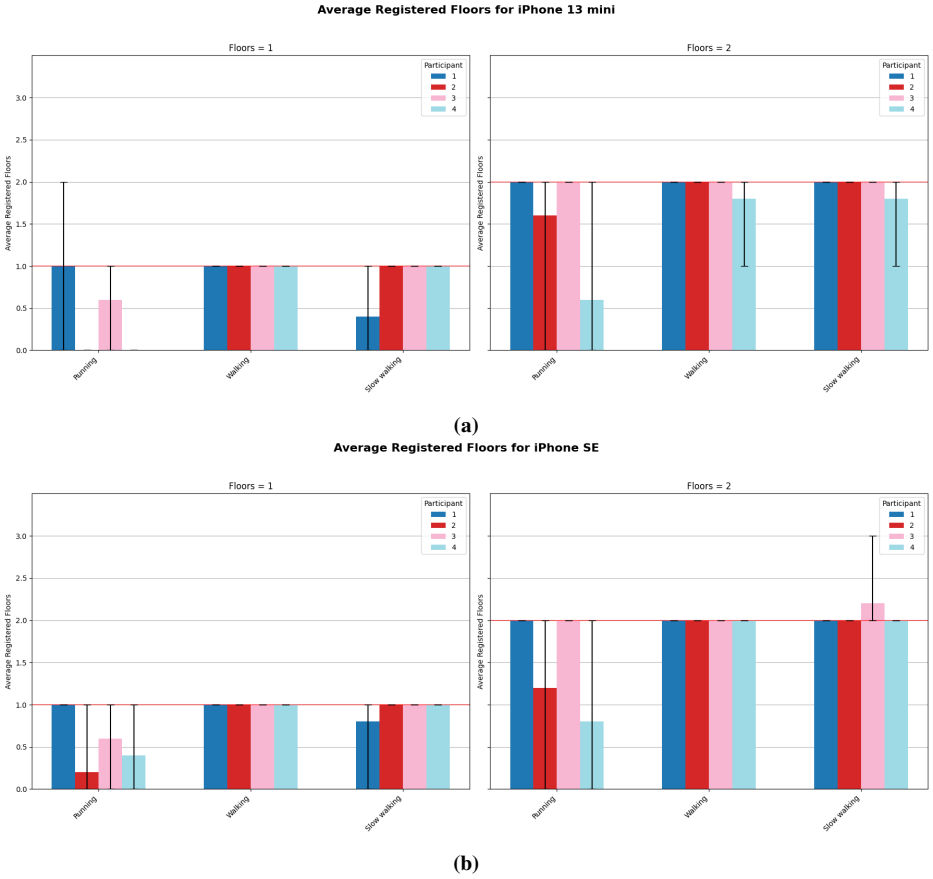


Fig. 6: Average registered floors when ascending for each condition and participants for iPhone 13m (a) and iPhone SE (b). The different conditions are shown on the x-axis (running, walking, slow-walking), and on the y-axis the average number of floors that were registered by the iPhone. Each participant is represented by a different colour. The error bars show the highest registered number of floors (top), and the lowest registered number of floors (bottom). The red dashed line represents the actual number of floors.

C. Floor data

1) Floor registration: An overview of the results for the floor registration can be found in Figure 6. The results for one and two floors are comparable for both phones.

For the walking condition, the iPhones were almost always correct in registering the number of floors. Only for participant 4 the iPhone 13m did not always correctly register two floors.

For the slow-walking condition, the iPhones displayed some error, registering no floors or only one in case of walking two floors. One time for participant 3 it registered three floors instead of the expected two.

When running, the iPhones were considerably worse in correct registration of floors. Only for participant 1, it almost always correctly registered the right number of floors. For the other participants, it often registered no floors, especially when walking one floor.

2) Floor accuracy: To determine how well the iPhones register floors when ascending, the accuracy of correctly registered floors was determined. The results for one floor, two floors, and the total accuracy score can be found in Table IV. For both iPhones, the phones registered floors most accurately when walking. The iPhone 13m had a total accuracy score of 97.5%, and the iPhone SE had a higher total accuracy score of 100%. Slow-walking led to a decrease in accuracy (total accuracy: iPhone 13m, 90%; iPhone SE, 95%), around 90% for both phones. When running the accuracy dropped even further, and gave the lowest accuracy scores of all conditions (total accuracy: iPhone 13m, 52.5%; iPhone SE, 62.5%). For the running condition the iPhones registered two floors more accurately than one floor.

TABLE IV: Accuracy scores for floor registration for walking one floor and two floors. The overall accuracy, combining the accuracy scores for both floors, is given by the total accuracy.

iPhone	Type	Total accuracy (%)	Total accuracy 1 floor (%)	Total accuracy 2 floors (%)
13m	slow walking	90	85	95
	walking	97.5	100	95
	running	52.5	30	75
SE	slow walking	95	95	95
	walking	100	100	100
	running	62.5	55	70

3) Floor accuracy with velocity: To further analyse the velocity and accuracy of floor registration, the average velocity of each participant was computed and plotted against the accuracy scores for one and two floors, seen in Figure 7. Between an average velocity of 0.60 m/s and 1.00 m/s the registration accuracy is the most accurate, often 100%. When going slower than this interval the accuracy slightly drops. When going faster than this interval, the higher the average velocity, the less accurate the registration. The right outermost data points correspond to the running trials of participant 2. These points do show a better accuracy for registering 2 floors, while the other data points for running show a continuous decrease in registration accuracy as the velocity increases.

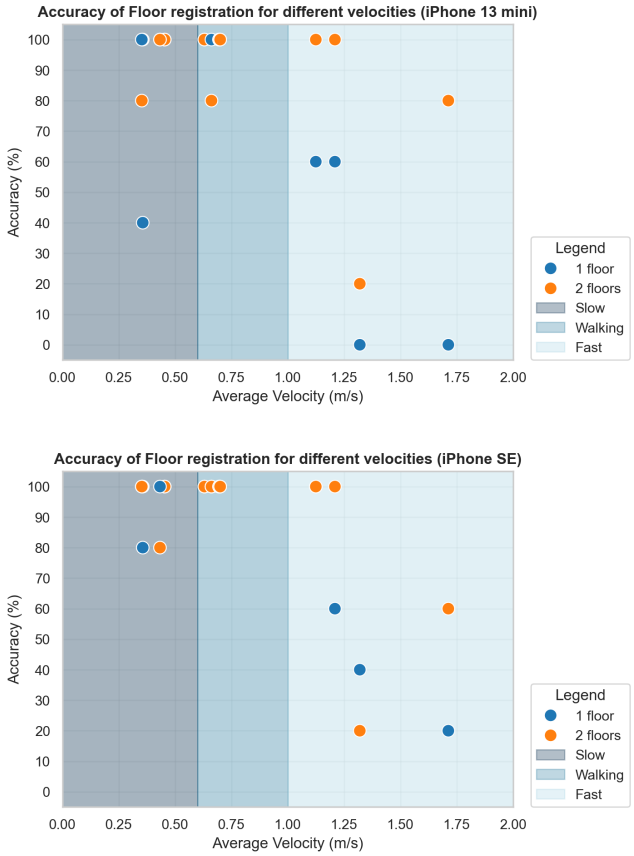


Fig. 7: Average velocities for each participant plotted against the accuracy scores for the iPhone 13m (A), and the iPhone SE (B). On the x-axis the average velocity is shown in meters per seconds, and on the y-axis the accuracy in percentages. The number of floors that were walked are indicated by the blue (1 floor) and orange (2 floors) dots. The different walking conditions are shown with the blue gradient. Each datapoint represents the velocities averaged over all trials of one participant in the respective walking condition.

D. Discussion

The aim of this research was to study the reliability of iPhones in registering flights climbed, by looking at the effect of three different walking speeds on the accuracy and error margin of step and floor registration using Apple's Health app. The results show that walking speed does influence correct registration of steps and floors. There is always some error, although its magnitude is dependent on walking speed. Errors are smallest during normal walking, increase when slow-walking, and are largest when running.

1) Results interpretation of step count registration: When looking at the results of the step count analysis it becomes clear that for the running condition the overall MAE showed the largest errors. Moreover, slow-walking showed greater registration errors than normal walking, though these were still smaller than the errors observed for running. A possible explanation for this could be that the step count registration on the Health app is more tailored towards normal walking. Therefore, changing the movement velocities could possibly influence the mechanisms and sensors that are responsible for the step registration. Unfortunately, Apple does not provide transparency about which specific sensors are involved or how the underlying mechanisms function in the iPhones,

including how the data is registered or processed. Therefore, it can only be guessed that the different velocities negatively influence the precision of the step registration.

Additionally, the trials included walking diagonal distances on stairs rather than on flat, horizontal surfaces. Previous research on overall step registration accuracy for walking set routes showed an average step accuracy with a MAPE of 1.9% to 2.3% for normal walking [4]. Although this metric differs from the MAE used in this study, the results for the walking condition during the diagonal stair trials indicate a similarly low level of error. This suggests that walking diagonally on stairs may not substantially affect step count accuracy.

Since the slow-walking velocities were closer to the normal walking velocity, this may help explain the larger error margins observed during running. Additionally, slow-walking more closely resembles normal walking in terms of the physical movements, which could further explain to the greater errors observed for running. During the slow-walking trials it was noticed that participants moved around more to find balance or moved more with their legs to carry the suitcase, this could explain the over-registering of the steps by the phones. Running tended to under-register the steps, which could be due to the quick movements, making it more difficult to distinguish separate steps. The physical differences between ascending and descending may also help explain the variation in error margins between the two conditions. Ascending generally requires more physical effort and deliberate movement when taking each step, whereas descending is often more about controlling and balancing yourself as you go down in a more fluid motion. This different kind of movement may lead to differences in detection, causing ascending to have a better precision than descending.

Finally, during the running trials, participant 2 took two steps at a time to move faster, as this was their normal habit when running up the stairs. This behavior was accounted for in the calculations and graphs. Interestingly, compared with the other participants, the average MAE for this participant's running trials was lower. This suggests that increasing both step length and velocity simultaneously may improve the precision of step registration.

2) Results interpretation of floor registration: Similar to the step count analysis, the overall accuracy of floor registration was better for normal walking speeds than for running or slow-walking. As mentioned earlier in the paper, the floor registration depends on a change in altitude and acceleration [2]. However, because of Apple's lack of transparency, the exact mechanisms and registration process remain unclear. Therefore, it can only be inferred that when running up a flight of stairs, some aspect of the sensors or mechanism is negatively influenced, reducing the overall precision of floor registration.

Interestingly, the accuracy for the two-floor trials was overall better for running and slow-walking than for the one-floor trials. One possible explanation is that a greater altitude change, and therefore a larger pressure difference, allowed for more accurate registration. Moreover, because these trials took longer to complete, the overall time to detect and register movements increased, potentially resulting in greater precision.

3) Combining steps and floors results: There were no noticeable differences in the step count and floor registration results. For both step and floor registration the HealthApp data shows less precision for the running and slow-walking condition. It occurred that the iPhones registered steps and no floors or they registered neither steps nor floors, however, they never recorded floors and no steps. This is in line with the knowledge that floor registration relies on movement, and both measurements use similar motion and sensor data, such as the accelerometer. For both, when motion patterns become irregular it reduces the reliability of both step and floor detection. However, a more thorough statistical analysis into the possible correlation should be performed with more data to provide a more concise and significant answer.

4) Limitations: There are several limitations to this study. Deciding on a number of trials for each condition that would ensure representative results, was greatly influenced by participant availability and imposed time constraints. Additionally, the lack of access to jailbroken phones limited the dataset to only the information provided in the exported file from the Health app. For analysis, it was impossible to obtain other data types such as the descended floors.

Furthermore, the use of two different iPhone models running separate iOS versions, combined with the exclusion of Android devices limits the understanding of algorithmic variance.The oversized suitcase, used to alter participants' gait, imposed significant physical constraints, especially for shorter individuals. This could have led to altered walking patterns, which could have influenced the results.

A previous study [4] highlighted the impact of phone carrying location on data registration, which could be further developed on our experimental model, as people tend to vary the location of their phone throughout daily activities.

From a statistical standpoint, the small sample size of participants restricted our ability to draw definitive conclusions on the cause of error in results. Participant 2 took two steps at a time when running, which posed a problem for the data-analysis of step registration, as the ground truth differed from the other participants. It is best to further standardise the experimental setup. Participant 2 also represented a significant outlier in subject measurements, due to a height difference compared to the other three participants.

Finally, software settings could have lead to skewed data, due to lack of biometric personalisation; because the Health app was not calibrated for each participant's specific age, sex, height, and weight, the software's internal algorithms likely relied on the existing preset for a female, aged 24, for the iPhone SE. For the iPhone 13m, the lack of presets potentially carried no effect on the gathered data. In reality, these factors could have little to no influence in criminal cases, since it is more likely for people to carry their personal device, but it is important to acknowledge this limitation for such studies. The final decision to keep the iPhones' display turned on at all times possibly affected how the Health app processed and timestamped activity data. Moreover, in real-life the Health app usually runs in the background, making the research findings less realistic.

E. Conclusion

In conclusion, this study provides a quantitative assessment of how walking speed affects the forensic reliability of Apple's Health app as a source of digital evidence for step and floor registration when climbing stairs. The findings show that regular walking speed presents low error margins, with a success rate of 97.5% to 100%. Variations in speed significantly alter the accuracy of data registration. Running produces the most errors, with floor registration dropping down to 52.5% and step-count consistently under-registering. Conversely, slow-walking, simulated by carrying a heavy object, led to step over-registration.

The findings highlight that walking speed influences correct registration by Apple's Health app, especially when running. Although, in the context of regular walking, floor registration data could prove useful in reconstructing events or verifying alibis. It should be noted that this data should not be used as evidence on its own, due to individuals and their daily behavioural habits being highly personalised. This can result in highly inaccurate registration and, therefore, lower reliability. Much more research has to be done to get a better understanding of the reliability of floor registration data.

1) Future work: As highlighted by the limitations presented above, there are many aspects of this research that could be addressed in future projects, to improve reliability and validity of the results. First, using the same phone model with different iOS versions could lead to a better understanding of iOS-specific inaccuracies and differences in system code. Second, expanding analysis to other data types besides steps and floors could provide more context for the sensor mechanisms. Third, expanding the subject pool would offer a great variety of heights and gait speeds, increasing experimental precision, bringing more statistical power, and neutralising outliers. The current significant challenge faced was the inability of obtaining jailbroken iPhones, not allowing for an in depth analysis of a more extensive dataset.

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F. Workload distribution

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Discussion	**Interpretations:** Yasmin Veldhuis, Marjolein van Tol **Limitations:** Mara Săcuiu, Napoleon van Assen
Conclusion	Yasmin Veldhuis, Mara Săcuiu, Napoleon van Assen
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